

JDBC CRUD Coding Questions with Solutions

Question 1: Insert Student Record Problem

Create a Java program using JDBC to insert a student record into MySQL database.

Table:

```
CREATE TABLE student(  
    id INT,  
    name VARCHAR(50)  
);
```

Solution

```
import java.sql.Connection;  
import java.sql.DriverManager;  
import java.sql.Statement;  
  
public class InsertStudent {  
  
    public static void main(String[] args) {  
  
        try {  
  
            // Load Driver  
            Class.forName("com.mysql.cj.jdbc.Driver");  
  
            // Create Connection  
            Connection con = DriverManager.getConnection(  
                "jdbc:mysql://localhost:3306/test",  
                "root",  
                "password"  
            );  
  
            // Create Statement  
            Statement st = con.createStatement();  
  
            // Insert Query  
            String query = "INSERT INTO student VALUES(1,'Ekta')";  
  
            // Execute Query  
            st.executeUpdate(query);  
  
            System.out.println("Data Inserted Successfully");  
  
            // Close Connection  
            con.close();  
  
        } catch (Exception e) {  
            System.out.println(e);  
        }  
    }  
}
```

Question 2: Display All Records (READ) Problem

Write JDBC program to fetch all student records.

Solution

```
import java.sql.Connection;  
import java.sql.DriverManager;  
import java.sql.ResultSet;  
import java.sql.Statement;  
  
public class FetchStudent {  
  
    public static void main(String[] args) {  
  
        try {  
  
            Class.forName("com.mysql.cj.jdbc.Driver");  
  
            Connection con = DriverManager.getConnection(  
                "jdbc:mysql://localhost:3306/test",  
                "root",  
                "password"  
            );  
  
        }  
    }  
}
```

```

        Statement st = con.createStatement();

        String query = "SELECT * FROM student";

        ResultSet rs = st.executeQuery(query);

        while(rs.next()) {

            System.out.println(
                rs.getInt("id") + " "
                + rs.getString("name")
            );

        }

        con.close();

    } catch(Exception e) {
        System.out.println(e);
    }
}

```

Question 3: Update Student Name
Problem

Update student name whose id = 1.

Solution

```

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.Statement;

public class UpdateStudent {

    public static void main(String[] args) {

        try {

            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/test",
                "root",
                "password"
            );

            Statement st = con.createStatement();

            String query =
                "UPDATE student SET name='Priya' WHERE id=1";

            st.executeUpdate(query);

            System.out.println("Data Updated Successfully");

            con.close();

        } catch(Exception e) {
            System.out.println(e);
        }

    }
}

```

Question 4: Delete Student Record
Problem

Delete student whose id = 1.

Solution

```

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.Statement;

public class DeleteStudent {

    public static void main(String[] args) {

        try {

            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(

```

```

        "jdbc:mysql://localhost:3306/test",
        "root",
        "password"
    );

    Statement st = con.createStatement();

    String query =
        "DELETE FROM student WHERE id=1";

    st.executeUpdate(query);

    System.out.println("Data Deleted Successfully");

    con.close();
} catch (Exception e) {
    System.out.println(e);
}
}
}

```

Question 5: Insert Data Using PreparedStatement
Problem

Insert user input dynamically using PreparedStatement.

Solution

```

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.util.Scanner;

public class InsertUsingPS {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        try {

            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/test",
                "root",
                "password"
            );

            String query =
                "INSERT INTO student VALUES(?, ?)";

            PreparedStatement ps =
                con.prepareStatement(query);

            System.out.print("Enter ID: ");
            int id = sc.nextInt();

            sc.nextLine();

            System.out.print("Enter Name: ");
            String name = sc.nextLine();

            ps.setInt(1, id);
            ps.setString(2, name);

            ps.executeUpdate();

            System.out.println("Data Inserted");

            con.close();

        } catch (Exception e) {
            System.out.println(e);
        }
    }
}

```